Lecture 9: Vector Autoregressive (VAR) Models

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Vector Autoregressive (VAR) Models

Introduction

Recap:

- Last class, we studied forecasting and impulse response functions.
- We focused on a univariate time-series model.
  - Intertemporal: $y_t \leftarrow y_{t-1}$
  - Policy effects propagate over time.

- In reality, multiple variables are determined jointly.
  - e.g., supply and demand; unemployment and production
  - Policy effects spillover to other economic variables.

- Today, we extend our analysis to a multivariate model.
  - Intertemporal: $y_t, x_t \leftarrow y_{t-1}, x_{t-1}$
  - Intratemporal: $y_t \leftrightarrow x_t$

Outline

- Preliminary: Linear Algebra
- Definition
- Identification
- Estimation and Inference
- Granger Causality

Preliminary

Linear Algebra

Definition: Vectorization

Let A be a $K \times L$ matrix:

$$A = \begin{bmatrix} a_{11} & \dots & a_{1L} \\ \vdots & \ddots & \vdots \\ a_{K1} & \dots & a_{KL} \end{bmatrix} = [a_{\cdot 1} \ \dots \ a_{\cdot L}],$$

where $a_{\cdot \ell} := [a_{1\ell} \ \dots \ a_{K\ell}]'$ for all $\ell = 1, \dots, L$. The vectorization of A, denoted by $vec(A)$, is defined as

$$vec(A) := \begin{bmatrix} a_{\cdot 1} \\ \vdots \\ a_{\cdot L} \end{bmatrix}.$$

- The $vec(\cdot)$ operator transforms a $K \times L$ matrix into a $KL \times 1$ vector.

Preliminary: Linear Algebra

Definition: Kronecker Product

Let A be a $K \times L$ matrix:

$$A = \begin{bmatrix} a_{11} & \dots & a_{1L} \\ \vdots & \ddots & \vdots \\ a_{K1} & \dots & a_{KL} \end{bmatrix}$$

Let B be an $M \times N$ matrix. The Kronecker product $A \otimes B$ is defined as

$$A \otimes B := \begin{bmatrix} a_{11}B & \dots & a_{1L}B \\ \vdots & \ddots & \vdots \\ a_{K1}B & \dots & a_{KL}B \end{bmatrix}$$

- The Kronecker product $\otimes$ combines to a $K \times L$ matrix and an $M \times N$ matrix into a $KM \times LN$ matrix.
- In general, $A \otimes B \neq B \otimes A$.

- Ex) Let A and B be 2×2 matrix:

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} 11 & 12 \\ 13 & 14 \end{bmatrix}.$$

  Vectorization:

$$vec(A) = \begin{bmatrix} 1 \\ 3 \\ 2 \\ 4 \end{bmatrix} \quad \text{and} \quad vec(B) = \begin{bmatrix} 11 \\ 13 \\ 12 \\ 14 \end{bmatrix}.$$

Preliminary: Linear Algebra

Kronecker product:

$$A \otimes B = \begin{bmatrix} 1 \begin{bmatrix} 11 & 12 \\ 13 & 14 \end{bmatrix} & 2 \begin{bmatrix} 11 & 12 \\ 13 & 14 \end{bmatrix} \\ \\ 3 \begin{bmatrix} 11 & 12 \\ 13 & 14 \end{bmatrix} & 4 \begin{bmatrix} 11 & 12 \\ 13 & 14 \end{bmatrix} \end{bmatrix} = \begin{bmatrix} 1 & 2 & 22 & 24 \\ 3 & 4 & 26 & 28 \\ 33 & 36 & 44 & 48 \\ 39 & 42 & 52 & 56 \end{bmatrix},$$

and

$$B \otimes A = \begin{bmatrix} 11 \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} & 12 \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \\ \\ 13 \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} & 14 \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \end{bmatrix} = \begin{bmatrix} 11 & 22 & 12 & 24 \\ 33 & 44 & 36 & 48 \\ 13 & 26 & 14 & 28 \\ 39 & 52 & 42 & 56 \end{bmatrix}.$$

Preliminary: Linear Algebra

Definition: Positive Semidefinite

A symmetric square matrix A is positive semidefinite if $x'Ax \geq 0$ for all column vectors x.

Definition: Positive Definite

A symmetric square matrix A is positive semidefinite if $x'Ax > 0$ for all column vectors x.

- In general, variance-covariance matrices are positive semidefinite, but not positive definite.

Preliminary: Linear Algebra

Proposition: Cholesky Decomposition

Let A be a positive definite matrix. Then, there exists a lower triangler matrix P such that

$$A = PP'.$$

When all the diagonal elements of P are positive, P is unique.

Proposition: Modified Cholesky Decomposition (LDL Decomposition)

Let A be a positive definite matrix. Then, there exists a lower triangler matrix L with one along the diagonal element and a diagonal matrix D such that

$$A = LDL'.$$

When all the diagonal elements of D are positive, L and D are unique.

- Clearly, $P = LD^{\frac{1}{2}}$.

- Ex) Cholesky decomposition:

$$\underbrace{\begin{bmatrix} 4 & -12 & 8 \\ -12 & 52 & -40 \\ 8 & -40 & 41 \end{bmatrix}}_{A} = \underbrace{\begin{bmatrix} 2 & 0 & 0 \\ -6 & 4 & 0 \\ 4 & -4 & 3 \end{bmatrix}}_{P} \underbrace{\begin{bmatrix} 2 & -6 & 4 \\ 0 & 4 & -4 \\ 0 & 0 & 3 \end{bmatrix}}_{P'}$$

- Ex) Modified Cholesky decomposition:

$$\underbrace{\begin{bmatrix} 4 & -12 & 8 \\ -12 & 52 & -40 \\ 8 & -40 & 41 \end{bmatrix}}_{A} = \underbrace{\begin{bmatrix} 1 & 0 & 0 \\ -3 & 1 & 0 \\ 2 & -1 & 1 \end{bmatrix}}_{L} \underbrace{\begin{bmatrix} 4 & 0 & 0 \\ 0 & 16 & 0 \\ 0 & 0 & 9 \end{bmatrix}}_{D} \underbrace{\begin{bmatrix} 1 & -3 & 2 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{bmatrix}}_{L'}$$

Vector Autoregressive (VAR) Models

Definition

Definition: The Object of Interest

- Consider a bivariate time-series data $\{(x_s, z_s)\}_{s=1}^{t}$.
  - e.g., x_t: quantity of supply, z_t: quantity of demand
- x_t and z_t are subject to "shocks" ε_t^x and ε_t^z, respectively.
  - e.g., ε_t^x: supply shocks, ε_t^z: demand shocks
- Denote $Y_t := [x_t \ z_t]'$, $\boldsymbol{\varepsilon}_t := [\varepsilon_t^x \ \varepsilon_t^z]'$, and $\mathcal{I}_t := \{(x_s, z_s)\}_{s=1}^{t}$.
- The object of interest:

$$
\begin{aligned}
IRF_t(h) &:= \frac{\partial E[Y_{t+h}|\mathcal{I}_t]}{\partial \boldsymbol{\varepsilon}_t} \\
&= \begin{bmatrix} \frac{\partial E[x_{t+h}|\mathcal{I}_t]}{\partial \varepsilon_t^x} & \frac{\partial E[z_{t+h}|\mathcal{I}_t]}{\partial \varepsilon_t^x} \\ \frac{\partial E[x_{t+h}|\mathcal{I}_t]}{\partial \varepsilon_t^z} & \frac{\partial E[z_{t+h}|\mathcal{I}_t]}{\partial \varepsilon_t^z} \end{bmatrix}.
\end{aligned}
$$

Definition: Structural Vector Autoregression Model

- We need a model to link (x_{t+h}, z_{t+h}) and $\mathcal{I}_t$.
  - intratemporal and intertemporal relationships

- Structural Vector Autoregression Model (SVAR):

$$x_t = \alpha_0 + \alpha_1 z_t + \alpha_2 x_{t-1} + \alpha_3 z_{t-1} + \varepsilon_t^x$$
$$z_t = \beta_0 + \beta_1 x_t + \beta_2 x_{t-1} + \beta_3 z_{t-1} + \varepsilon_t^x,$$

  where $\varepsilon_t^x \sim WN(0, \sigma^2_{\varepsilon^x})$ and $\varepsilon_t^z \sim WN(0, \sigma^2_{\varepsilon^z})$ with $Cov(\varepsilon_t^x, \varepsilon_t^z) = 0$.
- α_1, β_1: effects of a unit change in contemporaneous variables.
- α_2, α_3, β_2, β_3,: effects of a unit change in lagged variables.

- Usually, SVAR is directly implied by an economic theory.

Definition: Structural Vector Autoregression Model

- In matrix notation:

$$\begin{bmatrix} x_t \\ z_t \end{bmatrix} = \begin{bmatrix} \alpha_0 \\ \beta_0 \end{bmatrix} + \begin{bmatrix} 0 & \alpha_1 \\ \beta_1 & 0 \end{bmatrix} \begin{bmatrix} x_t \\ z_t \end{bmatrix} + \begin{bmatrix} \alpha_2 & \alpha_3 \\ \beta_2 & \beta_3 \end{bmatrix} \begin{bmatrix} x_{t-1} \\ z_{t-1} \end{bmatrix} + \begin{bmatrix} \varepsilon_t^x \\ \varepsilon_t^z \end{bmatrix}$$

$$\therefore \quad \underbrace{\begin{bmatrix} 1 & -\alpha_1 \\ -\beta_1 & 1 \end{bmatrix}}_{A_0} \underbrace{\begin{bmatrix} x_t \\ z_t \end{bmatrix}}_{Y_t} = \underbrace{\begin{bmatrix} \alpha_0 \\ \beta_0 \end{bmatrix}}_{A_1} + \underbrace{\begin{bmatrix} \alpha_2 & \alpha_3 \\ \beta_2 & \beta_3 \end{bmatrix}}_{A_2} \underbrace{\begin{bmatrix} x_{t-1} \\ z_{t-1} \end{bmatrix}}_{Y_{t-1}} + \underbrace{\begin{bmatrix} \varepsilon_t^x \\ \varepsilon_t^z \end{bmatrix}}_{\boldsymbol{\varepsilon}_t}$$

$$\therefore \quad A_0 Y_t = A_1 + A_2 Y_{t-1} + \boldsymbol{\varepsilon}_t,$$

where

$$\Xi_{\boldsymbol{\varepsilon}} := Var(\boldsymbol{\varepsilon}_t) = \begin{bmatrix} \sigma^2_{\varepsilon^x} & 0 \\ 0 & \sigma^2_{\varepsilon^z} \end{bmatrix}.$$

- A_0, A_1, A_2, and $\Xi_{\boldsymbol{\varepsilon}}$: structural parameters

Definition: Reduced-Form Vector Autoregression Model

- Reduced-form Vector Autoregressive Model (VAR):

$$Y_t = A_0^{-1} A_1 + A_0^{-1} A_2 Y_{t-1} + A_0^{-1} \varepsilon_t,$$

  provided A_0^{-1} exists.

- A_0^{-1} captures the infinite rounds of feedback between contemporaneous variables.

Definition: Reduced-Form Vector Autoregression Model

- Denote

$$B_0 := A_0^{-1} A_1, \qquad B_1 := A_0^{-1} A_2, \qquad \boldsymbol{\eta}_t = \begin{bmatrix} \eta_t^x \\ \eta_t^z \end{bmatrix} := A_0^{-1} \boldsymbol{\varepsilon}_t$$

- Then,

$$Y_t = B_0 + B_1 Y_{t-1} + \boldsymbol{\eta}_t$$

where

$$\Xi_{\boldsymbol{\eta}} := Var(\boldsymbol{\eta}_t) = A_0^{-1} \Xi_{\boldsymbol{\varepsilon}} (A_0^{-1})'.$$

- B_0, B_1, and $\Xi_{\boldsymbol{\eta}}$: reduced-form parameters

Definition: Reduced-Form Vector Autoregression Model

- The characteristic equation:

$$det(\mathbf{I}_2 - B_1 s^*) = 0,$$

  where s^* is a scalar satisfying this equation.
- Proposition: The reduced-form VAR(1) is weakly stationary if and only if $|s^*| > 1$.

Proposition: Stability of VAR(p) model

The time-sereis process $\{Y_t\}$ with the reduced-form n-variate VAR(p) model

$$Y_t = B_0 + B_1 Y_{t-1} + B_2 Y_{t-2} + \ldots + B_p Y_{t-p} + \boldsymbol{\eta}_t$$

is weakly stationary if and only if all the solution s^* for the characteristic equation

$$det(\mathbf{I}_n - B_1 s - B_2 s^2 - \ldots - B_p s^p) = 0$$

satisfy $|s^*| > 1$.

▸ Sketch of proof

Definition: The Object of Interest Redefined

- By repeated substitution,

$$
\begin{aligned}
&Y_t = B_0 + B_1 Y_{t-1} + \boldsymbol{\eta}_t \\
&\implies Y_{t+h} = B_0 \sum_{j=0}^{h-1} B_1^j + B_1^{h+1} Y_{t-1} + B_1^h \underbrace{\boldsymbol{\eta}_t}_{A_0^{-1}\boldsymbol{\varepsilon}_t} + \sum_{j=0}^{h-1} B_1^j \underbrace{\boldsymbol{\eta}_{t+h-j}}_{A_0^{-1}\boldsymbol{\varepsilon}_{t+h-j}} .
\end{aligned}
$$

- $E[Y_{t+h}|\mathcal{I}_{t-1} \cup \{\varepsilon_t\}] = B_0 \sum_{j=0}^{h-1} B_1^j + B_1^{h+1} Y_{t-1} + B_1^h A_0^{-1} \varepsilon_t$
- $E[Y_{t+h}|\mathcal{I}_{t-1} \cup \{\varepsilon_t + \delta\}] = B_0 \sum_{j=0}^{h-1} B_1^j + B_1^{h+1} Y_{t-1} + B_1^h A_0^{-1} (\varepsilon_t + \delta)$

Hence,

$$IRF_t(h) = B_1^h A_0^{-1}.$$

- To identify $IRF_t(h)$ requires the identification of A_0 and B_1.

Vector Autoregressive (VAR) Models

Identification

Identification: Overview

- We want to recover A_0 and B_1.
- Our approach proceeds in two steps:
  1. Reduced-form parameter B_1.
  2. Structural parameter A_0.
- Identifying the strutural parameter A_0 requires additional assumptions.

Identification: Reduced-Form Parameters

- The reduced-form VAR can be written as

$$\begin{aligned} Y_t &= B_0 + B_1 Y_{t-1} + \boldsymbol{\eta}_t \\ &= \underbrace{\begin{bmatrix} B_0 & B_1 \end{bmatrix}}_{B'} \underbrace{\begin{bmatrix} 1 \\ Y_{t-1} \end{bmatrix}}_{X_t} + \boldsymbol{\eta}_t \\ &= B' X_t + \boldsymbol{\eta}_t, \end{aligned}$$

where

$$\Xi_{\boldsymbol{\eta}} := Var(\boldsymbol{\eta}_t) = A_0^{-1} \Xi_{\boldsymbol{\varepsilon}} (A_0^{-1})'.$$

- This looks very much the same as the standard AR(1) model!

Identification: Reduced-Form Parameters

Assumption 1 (TS1'): Linearity

A weakly stationary and weakly dependent stochastic process $\{Y_s, X_s\}$ follows

$$Y_t = B'X_t + \boldsymbol{\eta}_t,$$

where $\{\boldsymbol{\eta}_t\}$ is a sequence of reduced-form errors.

- By construction,

$$B = \begin{bmatrix} B_0 & B_1 \end{bmatrix}' = \begin{bmatrix} b_{0,1} & b_{1,1} & b_{1,2} \\ b_{0,2} & b_{1,3} & b_{1,4} \end{bmatrix}' = \begin{bmatrix} b_{0,1} & b_{0,2} \\ b_{1,1} & b_{1,3} \\ b_{1,2} & b_{1,4} \end{bmatrix}$$

Identification: Reduced-Form Parameters

Assumption 2 (TS2'): No Perfect Collinearity

None of the explanatory variables is a perfect linear combination of the others.

Assumption 3 (TS3'): Contemporaneous Exogeneity

$E[\boldsymbol{\eta}_t|X_t] = \mathbf{0}$ for all t.

- $E[\boldsymbol{\eta}_t|X_t] = E[A_0^{-1}\boldsymbol{\varepsilon}_t|X_t] = A_0^{-1}E[\boldsymbol{\varepsilon}_t|X_t]$.

Identification: Reduced-Form Parameters

Proposition: Identification of B

Under TS1'–TS3', the reduced-form parameter matrix B is identified.

Proof. Taking transposes of TS1',

$$Y_t' = X_t' B + \boldsymbol{\eta}_t'.$$

Premultiplying X_t and taking expectation,

$$E[X_t Y_t'] = E[X_t X_t'] B + \underbrace{E[X_t \boldsymbol{\eta}_t']}_{\mathbf{0} \text{ by TS3'}}.$$

In view of TS2', $(E[X_t X_t'])^{-1}$ exists, so that

$$B = (E[X_t X_t'])^{-1} E[X_t Y_t'].$$

□

Identification: Reduced-Form Parameters

Corollary: Identification

Under TS1'–TS3', the reduced-form variance-covariance matrix $\Xi_{\boldsymbol{\eta}}$ is identified.

Proof. Plugging the identified B back into TS1',

$$\boldsymbol{\eta}_t = Y_t - B' X_t,$$

so that

$$\Xi_{\boldsymbol{\eta}} = E[(\boldsymbol{\eta}_t - E[\boldsymbol{\eta}_t])(\boldsymbol{\eta}_t - E[\boldsymbol{\eta}_t])']$$

is identified. □

Identification: Structural Parameters

- Q. Can we recover the structural parameters as well?
- There are 9 reduced-form parameters:

$$B_0 := A_0^{-1} A_1 = \begin{bmatrix} b_{0,1} \\ b_{0,2} \end{bmatrix}, \qquad B_1 := A_0^{-1} A_2 = \begin{bmatrix} b_{1,1} & b_{1,2} \\ b_{1,3} & b_{1,4} \end{bmatrix}, \qquad \Xi_{\boldsymbol{\eta}} = \begin{bmatrix} \sigma^2_{\eta^x_t} & \sigma_{\eta^x_t \eta^z_t} \\ \sigma_{\eta^x_t \eta^z_t} & \sigma^2_{\eta^z_t} \end{bmatrix}$$

- There are 10 structural parameters:

$$A_0 = \begin{bmatrix} 1 & -\alpha_1 \\ -\beta_1 & 1 \end{bmatrix}, \qquad A_1 = \begin{bmatrix} \alpha_0 \\ \beta_0 \end{bmatrix}, \qquad A_2 = \begin{bmatrix} \alpha_2 & \alpha_3 \\ \beta_2 & \beta_3 \end{bmatrix}, \qquad \Xi_{\boldsymbol{\varepsilon}} = \begin{bmatrix} \sigma^2_{\varepsilon^x_t} & 0 \\ 0 & \sigma^2_{\varepsilon^z_t} \end{bmatrix}$$

- Structural parameters are not identified...
  $\Longrightarrow$ One additional constraint is necessary.

Identification: Structural Parameters

- In general, for n-variate VAR models:

$$\begin{cases} B_0 : n \text{ parameters} \\ B_1 : n^2 \text{ parameters} \\ \Xi_{\boldsymbol{\eta}} : \frac{n(n+1)}{2} \text{ parameters} \end{cases} \qquad \begin{cases} A_0 : n(n-1) \text{ parameters} \\ A_1 : n \text{ parameters} \\ A_2 : n^2 \text{ parameters} \\ \Xi_{\boldsymbol{\varepsilon}} : n \text{ parameters} \end{cases}$$

$\hookrightarrow \frac{3}{2}n(n+1)$ redueced-form parameters $\qquad \hookrightarrow 2n^2 + n$ structural parameters

- Identification of strutural paramters requres $\frac{1}{2}n(n-1)$ additional restrictions.

- Q. What kinds of additional assumptions are needed to identify the structural parameters?

Identification: Structural Parameters

- There have been multiple approaches:
  1. Normalization
  2. Short-run restrictions
  3. Long-run restrictions
  4. Sign restrictions
  5. Heteroskedasticity
  6. Direct measurement
  5. Method of external instruments (a.k.a. proxy VAR method)

Identification: Structural Parameters: Normalization

- The variance-covariance matrix:

$$\underbrace{\begin{bmatrix} \sigma^2_{\eta^x_t} & \sigma_{\eta^x_t \eta^z_t} \\ \sigma_{\eta^x_t \eta^z_t} & \sigma^2_{\eta^z_t} \end{bmatrix}}_{\Xi_{\boldsymbol{\eta}}} = \underbrace{\begin{bmatrix} 1 & -\alpha_1 \\ -\beta_1 & 1 \end{bmatrix}^{-1}}_{A_0^{-1}} \underbrace{\begin{bmatrix} \sigma^2_{\varepsilon^x_t} & 0 \\ 0 & \sigma^2_{\varepsilon^z_t} \end{bmatrix}}_{\Xi_{\boldsymbol{\varepsilon}}} \underbrace{\begin{bmatrix} 1 & -\alpha_1 \\ -\beta_1 & 1 \end{bmatrix}^{-1'}}_{(A_0^{-1})'}$$

- By setting $\sigma^2_{\varepsilon^x_t} = 1$,
  - HLS: 3 reduced-form parameters (i.e., $\sigma^2_{\eta^x_t}$, $\sigma^2_{\eta^z_t}$, and $\sigma_{\eta^x_t \eta^z_t}$)
  - RHS: 3 structural parameters (i.e., α_1, β_1, and $\sigma^2_{\varepsilon^z_t}$)

    $\Longrightarrow$ The structural parameter matrix A_0 is identified.

- This approach is sufficient only up until three-variable cases...
  - $n \lesseqgtr \frac{1}{2} n(n-1)$

Identification: Structural Parameters: Short-Run Restrictions

- The variance-covariance matrix:

$$\underbrace{\begin{bmatrix} \sigma^2_{\eta^x_t} & \sigma_{\eta^x_t \eta^z_t} \\ \sigma_{\eta^x_t \eta^z_t} & \sigma^2_{\eta^z_t} \end{bmatrix}}_{\Xi_{\boldsymbol{\eta}}} = \underbrace{\begin{bmatrix} 1 & {\color{red}-\alpha_1} \\ -\beta_1 & 1 \end{bmatrix}^{-1}}_{A_0^{-1}} \underbrace{\begin{bmatrix} \sigma^2_{\varepsilon^x_t} & 0 \\ 0 & \sigma^2_{\varepsilon^z_t} \end{bmatrix}}_{\Xi_{\boldsymbol{\varepsilon}}} \underbrace{\begin{bmatrix} 1 & {\color{red}-\alpha_1} \\ -\beta_1 & 1 \end{bmatrix}^{-1\prime}}_{(A_0^{-1})'}$$

- By setting ${\color{red}-\alpha_1 = 0}$,
  - HLS: 3 reduced-form parameters (i.e., $\sigma^2_{\eta^x_t}$, $\sigma^2_{\eta^z_t}$, and $\sigma_{\eta^x_t \eta^z_t}$)
  - RHS: 3 structural parameters (i.e., β_1, $\sigma^2_{\varepsilon^x_t}$, and $\sigma^2_{\varepsilon^z_t}$)

    $\Longrightarrow$ The structural parameter matrix A_0 is identified.

Identification: Structural Parameters: Short-Run Restrictions

- Under this assumption,

$$\Xi_{\boldsymbol{\eta}} = A_0^{-1}\Xi_{\boldsymbol{\varepsilon}}(A_0^{-1})' \quad \Longrightarrow \quad \begin{cases} \sigma^2_{\eta^x_t} = \sigma^2_{\varepsilon^x_t} + \beta_1^2\sigma^2_{\varepsilon^z_t} \\ \sigma_{\eta^x_t\eta^z_t} = \beta_1\sigma^2_{\varepsilon^z_t} \\ \sigma^2_{\eta^z_t} = \sigma^2_{\varepsilon^z_t} \end{cases}$$

This can be soloved as

$$\begin{cases} \sigma^2_{\varepsilon^x_t} = \sigma^2_{\eta^x_t} - (\frac{\sigma_{\eta^x_t\eta^z_t}}{\sigma^2_{\eta^z_t}})^2\sigma^2_{\eta^z_t} \\ \sigma^2_{\varepsilon^z_t} = \sigma^2_{\eta^z_t} \\ \beta_1 = \frac{\sigma_{\eta^x_t\eta^z_t}}{\sigma^2_{\eta^z_t}} \end{cases}$$

$\hookrightarrow$ A_0 and $\Xi_{\boldsymbol{\varepsilon}}$ are identified.

$\hookrightarrow$ $A_1 = A_0B_0$ and $A_2 = A_0B_1$ are identified.

Identification: Structural Parameters: Short-Run Restrictions

- In general, making A_0 a lower triangular matrix imposes $\frac{n(n-1)}{2}$ restrictions ($\leftarrow$ good!)
- Mathematical facts:
  1. A_0^{-1} is lower triangular, too;
  2. $\Xi_{\boldsymbol{\varepsilon}}$ is a diagonal matrix;
  3. $\Xi_{\boldsymbol{\eta}}$ is positive definite.
- In practice, LDL decomposition of $\Xi_{\boldsymbol{\eta}}$ gives us A_0 and $\Xi_{\boldsymbol{\varepsilon}}$:

$$\Xi_{\boldsymbol{\eta}} = \underbrace{A_0^{-1}}_{L} \underbrace{\Xi_{\boldsymbol{\varepsilon}}}_{D} \underbrace{(A_0^{-1})'}_{L'}.$$

Identification: Structural Parameters: Short-Run Restrictions

- "A_0 is lower triangular" restricts the contemporaneous (or short-run) relationship:

  - in terms of variables:

$$\begin{bmatrix} x_t \\ z_t \end{bmatrix} = \begin{bmatrix} \alpha_0 \\ \beta_0 \end{bmatrix} + \begin{bmatrix} 0 & \color{red}{0} \\ \beta_1 & 0 \end{bmatrix} \begin{bmatrix} x_t \\ z_t \end{bmatrix} + \begin{bmatrix} \alpha_2 & \alpha_3 \\ \beta_2 & \beta_3 \end{bmatrix} \begin{bmatrix} x_{t-1} \\ z_{t-1} \end{bmatrix} + \begin{bmatrix} \varepsilon_t^x \\ \varepsilon_t^z \end{bmatrix}$$

  - in terms of shocks:

$$\begin{bmatrix} \eta_t^x \\ \eta_t^z \end{bmatrix} = \begin{bmatrix} 1 & \color{red}{0} \\ \beta_1 & 1 \end{bmatrix} \begin{bmatrix} \varepsilon_t^x \\ \varepsilon_t^z \end{bmatrix}$$

- The ordering is rationalized by institutional knowledge and timing assumptions.

- e.g., Sims (1980); Christiano et al. (1999); Bernanke et al. (2005)

Identification: Structural Parameters: Long-Run Restrictions

- Economic theory may have implications for the long-run effects.
- The long-run impulse response function:

$$LRIRF_t := \lim_{h\to\infty} \sum_{\ell=1}^{h} IRF_t(\ell) = (I - B_1)^{-1} A_0^{-1}.$$

- Ex) Blanchard and Quah (1989)
  - x_t: GDP growth, z_t: Unemployment
  - ε_t^x: supply shocks, ε_t^z: demand shocks
  - Idea: Demand shocks ε_t^z only have a temporary impact on GDP growth x_t.

  $\Longrightarrow$ The (1,2) entry of $LRIRF_t$ has to be 0.

Identification: Structural Parameters: Long-Run Restrictions

- In practice,
  - the economic theory is "designed" so that $LRIRF_t$ is lower triangular.
  - $\Xi_{\boldsymbol{\varepsilon}} = \mathbf{I}$ (i.e., normalization).
- Hence,

$$LRIRF_t(LRIRF_t)' = (\mathbf{I} - B_1)^{-1} \underbrace{A_0^{-1} A_0^{-1\prime}}_{\Xi_{\boldsymbol{\eta}}} (\mathbf{I} - B_1)^{-1\prime} = (\mathbf{I} - B_1)^{-1} \Xi_{\boldsymbol{\eta}} (\mathbf{I} - B_1)^{-1\prime}.$$

- The Cholesky decomposition of the RHS recovers $LRIRF_t$.

  $\Longrightarrow$ This in turn identifies $A_0 = (LRIRF_t)^{-1}(\mathbf{I} - B_1)^{-1}$.
- e.g., Blanchard and Quah (1989); King et al. (1991); Galí (1999)

Identification: Structural Parameters: Sign Restrictions (Optional)

- Economic theory may have implications (only) for the signs.
- Ex) Supply (x) and demand (z):
  - ε_t^x: supply shocks, ε_t^z: demand shocks
  - η_t^x: quantity shocks, η_t^z: price shocks
  - A supply shock ε_t^x will increase quantity η_t^x and reduce price η_t^z.
  - A demand shock ε_t^z will increase both quantity η_t^x and price η_t^z.

$$\begin{bmatrix} \eta_t^x \\ \eta_t^z \end{bmatrix} = \begin{bmatrix} + & + \\ - & + \end{bmatrix} \begin{bmatrix} \varepsilon_t^x \\ \varepsilon_t^z \end{bmatrix}$$

- This approach only recovers the set of possible combinations (i.e., set identification).
- e.g., Faust (1998); Canova and De Nicoló (2002); Uhlig (2005)

Identification: Structural Parameters: Heteroskedasticity (Optional)

Identification by regime-shift heteroskedasticity:

- A_0: constant over the full sample period.
- Two variance regimes: Regime 1 ($\Xi_{\boldsymbol{\varepsilon}}^1$ and $\Xi_{\boldsymbol{\eta}}^1$) and Regime 2 ($\Xi_{\boldsymbol{\varepsilon}}^2$ and $\Xi_{\boldsymbol{\eta}}^2$).
- The variance-covariance matrix:

$$\begin{cases} \Xi_{\boldsymbol{\eta}}^1 = A_0^{-1} \Xi_{\boldsymbol{\varepsilon}}^1 A_0^{-1\prime} \\ \Xi_{\boldsymbol{\eta}}^2 = A_0^{-1} \Xi_{\boldsymbol{\varepsilon}}^2 A_0^{-1\prime} \end{cases}$$

  LHS: $\frac{n(n+1)}{2} \times 2 = n(n+1)$ parameters

  RHS: $n(n-1) + n \times 2 = n(n+1)$ parameters

  $\Longrightarrow A_0$ is identified!

- e.g., Rigobon (2003); Rigobon and Sack (2003, 2004)

Identification: Structural Parameters: Heteroskedasticity (Optional)

Identification by conditional heteroskedasticity:

- A_0: constant over the full sample period.
- ε_t: conditionally heteroskedastic.
- The variance-covariance matrix:

$$E[\boldsymbol{\eta}_t \boldsymbol{\eta}_t' | Y_{t-1}, Y_{t-2}, \ldots] = A_0^{-1} E[\varepsilon_t \varepsilon_t' | Y_{t-1}, Y_{t-2}, \ldots] A_0^{-1\prime}$$

- Devise a (G)ARCH model for ε_t.
  $\hookrightarrow$ $\boldsymbol{\eta}_t$ also follows a conditionally heteroskedastic process.
- With this model, A_0 can be identified.
- e.g., Sentana and Fiorentini (2001); Normandin and Phaneuf (2004)

Identification: Structural Parameters: Direct Measurement (Optional)

- This approach directly measures the structural shock of interest.
- Narrative approach:

  $\hookrightarrow$ Historical documents can be used to construct an additional time series.

  e.g., monetary shock series based on FOMC minutes;

  e.g., Ramey and Shapiro (1998); Ramey (2011); Romer and Romer (2004, 2010)

- High-frequency financial data:

  $\hookrightarrow$ High-frequency data are useful for capture unanticipated shocks.

  e.g., Kuttner (2001); Cochrane and Piazzesi (2002); Faust et al. (2004); Piazzesi and Swanson (2008); Gertler and Karadi (2015); Nakamura and Steinsson (2018)

Identification: Structural Parameters: Method of External Instruments (Optional)

- This approach uses information obtained "outside" the VAR model under study.
  - e.g., time-series based on narrative evidence;
  - e.g., shocks from estimated DSGE models.
- Such external series are noisy measures of the true shock.
  $\Longrightarrow$ can be used as instrumental variables.
- e.g., Stock and Watson (2012); Mertens and Ravn (2013); Gertler and Karadi (2015)

Vector Autoregressive (VAR) Models

Estimation and Inference

Estimation and Inference: Reduced-Form Parameters

- Given the identification result:

$$B = (E[X_t X_t'])^{-1} E[X_t Y_t'],$$

a natural estimator is the sample-analog:

$$\widehat{B} := \left(\frac{1}{T}\sum_{t=1}^{T} X_t X_t'\right)^{-1}\left(\frac{1}{T}\sum_{t=1}^{T} X_t Y_t'\right)$$

Proposition: Consistency

Under TS1'–TS3', $\widehat{B}$ is consistent for B, i.e., $\widehat{B} \xrightarrow{p} B$.

Estimation and Inference: Reduced-Form Parameters

- Define compact expressions for B an $\widehat{B}$:

$$B = \begin{bmatrix} b_{0,1} & b_{0,2} \\ b_{1,1} & b_{1,3} \\ b_{1,2} & b_{1,4} \end{bmatrix} = \begin{bmatrix} b_1 & b_2 \end{bmatrix}, \quad \text{where} \quad b_1 = \begin{bmatrix} b_{0,1} \\ b_{1,1} \\ b_{1,2} \end{bmatrix} \quad and \quad b_2 = \begin{bmatrix} b_{0,2} \\ b_{1,3} \\ b_{1,4} \end{bmatrix},$$

$$\widehat{B} = \begin{bmatrix} \widehat{b}_{0,1} & \widehat{b}_{0,2} \\ \widehat{b}_{1,1} & \widehat{b}_{1,3} \\ \widehat{b}_{1,2} & \widehat{b}_{1,4} \end{bmatrix} = \begin{bmatrix} \widehat{b}_1 & \widehat{b}_2 \end{bmatrix}, \quad \text{where} \quad \widehat{b}_1 = \begin{bmatrix} \widehat{b}_{0,1} \\ \widehat{b}_{1,1} \\ \widehat{b}_{1,2} \end{bmatrix} \quad and \quad \widehat{b}_2 = \begin{bmatrix} \widehat{b}_{0,2} \\ \widehat{b}_{1,3} \\ \widehat{b}_{1,4} \end{bmatrix}.$$

- Vectorize B and $\widehat{B}$:

$$b := vec(B) = \begin{bmatrix} b_1 \\ b_2 \end{bmatrix} \quad \text{and} \quad \widehat{b} := vec(\widehat{B}) = \begin{bmatrix} \widehat{b}_1 \\ \widehat{b}_2 \end{bmatrix}.$$

Estimation and Inference: Reduced-Form Parameters

Proposition: Asymptotic Normality of $\widehat{b}$

Under TS1'–TS3',

$$\sqrt{T}(\widehat{b} - b) \xrightarrow{d} \mathcal{N}(0, AVar(\widehat{b})),$$

where $AVar(\widehat{b}) := (\mathbf{I} \otimes E[X_t X_t']^{-1})\Omega(\mathbf{I} \otimes E[X_t X_t']^{-1})$ with $\Omega := \sum_{\ell=-\infty}^{\infty} E[\boldsymbol{\eta}_{t-\ell}\boldsymbol{\eta}_t \otimes X_{t-\ell}X_t]$.

- The shape of Ω changes depending on the assumptions about $\boldsymbol{\eta}_t$:

  Scenario 1: Homoskedasticity & No serial correlation

  Scenario 2: Heteroskedasticity & No serial correlation

  Scenario 3: Heteroskedasticity & Serial correlation

Estimation and Inference: Reduced-Form Parameters

- **Scenario 1**: Homoskedasticity & No serial correlation

$$\begin{aligned}\Omega &= E[\boldsymbol{\eta}_t \boldsymbol{\eta}_t \otimes X_t X_t] \\ &= \Sigma \otimes E[X_t X_t'], \quad \text{with} \quad \Sigma := E[\boldsymbol{\eta}_t \boldsymbol{\eta}_t' | X_t],\end{aligned}$$

so that

$$AVar(\widehat{b}) = \Sigma \otimes E[X_t X_t']^{-1}.$$

The variance estimator is

$$\widehat{\Omega} := \Big(\frac{1}{T}\sum_{t=1}^{T} \widehat{\boldsymbol{\eta}}_t \widehat{\boldsymbol{\eta}}_t'\Big) \otimes \Big(\frac{1}{T}\sum_{t=1}^{T} X_t X_t'\Big),$$

thereby

$$\widehat{AVar}(\widehat{b}) = \Big(\frac{1}{T}\sum_{t=1}^{T} \widehat{\boldsymbol{\eta}}_t \widehat{\boldsymbol{\eta}}_t'\Big) \otimes \Big(\frac{1}{T}\sum_{t=1}^{T} X_t X_t'\Big)^{-1}.$$

Estimation and Inference: Reduced-Form Parameters

- **Scenario 2**: Heteroskedasticity & No serial correlation

$$\Omega = \sum_{t=1}^{T} E[\boldsymbol{\eta}_t \boldsymbol{\eta}_t' \otimes X_t X_t']$$

The heteroskedasticity-robust variance estimator is

$$\widehat{\Omega} = \frac{1}{T} \sum_{t=1}^{T} \widehat{\boldsymbol{\eta}}_t \widehat{\boldsymbol{\eta}}_t' \otimes X_t X_t',$$

thereby

$$\widehat{AVar}(\widehat{b}) := \left(\mathbf{I} \otimes \left(\frac{1}{T} \sum_{t=1}^{T} X_t X_t' \right)^{-1} \right) \widehat{\Omega} \left(\mathbf{I} \otimes \left(\frac{1}{T} \sum_{t=1}^{T} X_t X_t' \right)^{-1} \right).$$

Estimation and Inference: Reduced-Form Parameters

- **Scenario 3**: Heteroskedasticity & Serial correlation

$$\Omega := \sum_{\ell=-\infty}^{\infty} E[\boldsymbol{\eta}_{t-\ell}\boldsymbol{\eta}_t \otimes X_{t-\ell}X_t].$$

The Newey-West estimator is

$$\widehat{\Omega}_M := \sum_{\ell=-M}^{\ell=M}\left(1-\frac{|\ell|}{M+1}\right)\frac{1}{T}\sum_{1\leq t-\ell\leq T}(\widehat{\boldsymbol{\eta}}_{t-\ell}\otimes X_{t-\ell})(\widehat{\boldsymbol{\eta}}_t'\otimes X_t'),$$

thereby

$$\widehat{AVar}(\widehat{b}) := \left(\mathbf{I}\otimes\left(\frac{1}{T}\sum_{t=1}^{T}X_tX_t'\right)^{-1}\right)\widehat{\Omega}_M\left(\mathbf{I}\otimes\left(\frac{1}{T}\sum_{t=1}^{T}X_tX_t'\right)^{-1}\right).$$

Estimation and Inference: Structural Parameters

- Define compact expressions for A and $\widehat{A}$:

$$A = \begin{bmatrix} a_1 & a_2 \end{bmatrix}, \quad \text{where} \quad a_1 := \begin{bmatrix} \alpha_0 \\ \alpha_1 \\ \alpha_2 \\ \alpha_3 \end{bmatrix} \quad \text{and} \quad a_2 := \begin{bmatrix} \beta_0 \\ \beta_1 \\ \beta_2 \\ \beta_3 \end{bmatrix};$$

$$\widehat{A} = \begin{bmatrix} \widehat{a}_1 & \widehat{a}_2 \end{bmatrix}, \quad \text{where} \quad \widehat{a}_1 := \begin{bmatrix} \widehat{\alpha}_0 \\ \widehat{\alpha}_1 \\ \widehat{\alpha}_2 \\ \widehat{\alpha}_3 \end{bmatrix} \quad \text{and} \quad \widehat{a}_2 := \begin{bmatrix} \widehat{\beta}_0 \\ \widehat{\beta}_1 \\ \widehat{\beta}_2 \\ \widehat{\beta}_3 \end{bmatrix}.$$

- Vectorize A and $\widehat{A}$:

$$a := vec(A) = \begin{bmatrix} a_1 \\ a_2 \end{bmatrix} \quad \text{and} \quad \widehat{a} := vec(\widehat{A}) = \begin{bmatrix} \widehat{a}_1 \\ \widehat{a}_2 \end{bmatrix}.$$

Estimation and Inference: Structural Parameters

- The identification result implies

$$a = g(b),$$

  where $g(\cdot)$ is a continuous map.

- The sample-analog estimator for a:

$$\widehat{a} := g(\widehat{b}).$$

Estimation and Inference: Structural Parameters

Proposition: Consistency of $\widehat{a}$

Under TS1'–TS3', $\widehat{a}$ is consistent for a, i.e., $\widehat{a} \xrightarrow{p} a$.

Sketch of proof. Use the continuous mapping theorem:

$$\widehat{b} \xrightarrow{p} b$$

$$\Downarrow$$

$$\underbrace{g(\widehat{b})}_{\widehat{a}} \xrightarrow{p} \underbrace{g(b)}_{a}$$

□

Estimation and Inference: Structural Parameters

Proposition: Asymptotic Distribution of $\widehat{a}$

Under TS1'–TS3',

$$\sqrt{T}(\widehat{a} - a) \xrightarrow{d} \mathcal{N}(0, G(b)AVar(\widehat{b})G(b)'),$$

where $G(b) := \frac{\partial g(b)}{\partial b}$.

Sketch of proof. Use the delta method:

$$\sqrt{T}(\widehat{b} - b) \xrightarrow{d} \mathcal{N}(0, AVar(\widehat{b}))$$

$$\Downarrow$$

$$\sqrt{T}(\underbrace{g(\widehat{b})}_{\widehat{a}} - \underbrace{g(b)}_{a}) \xrightarrow{d} \mathcal{N}(0, \left(\frac{\partial g(b)}{\partial b}\right) AVar(\widehat{b}) \left(\frac{\partial g(b)}{\partial b}\right)')$$

$\square$

Estimation and Inference: Impulse Response Functions

- The object of interest:
$$IRF_t(h) = B_1^h A_0^{-1}.$$
- Having $\widehat{A}_0$ and $\widehat{B}_1$ obtained, the sample-analog estimator for $IRF_t(h)$ is
$$\widehat{IRF}_t(h) = \widehat{B}_1^h \widehat{A}_0^{-1}.$$

- In a large sample, $\widehat{IRF}_t(h)$ is consistent for $IRF_t(h)$ and normally distributed.
- In a finite sample, $\widehat{IRF}_t(h)$ is not unbiased for $IRF_t(h)$ nor not normally distributed.

  $\hookrightarrow$ The bias tends to be large as h becomes large.

Vector Autoregressive (VAR) Models

Granger Causality

Granger Causality: Definition

Definition: Granger Causality (Granger, 1969; Sims, 1972)

For two time-series processes $\{(x_t, z_t)\}_{t=-\infty}^{\infty}$, we say that $\{z_t\}$ Granger-causes $\{x_t\}$ if

$$E[x_t | x_{t-1}, x_{t-2}, \dots, z_{t-1}, z_{t-2}, \dots] \neq E[x_t | x_{t-1}, x_{t-2}, \dots],$$

i.e., if the past values of $\{z_t\}$ help to forecast $\{x_t\}$.

- Let $\mathcal{I}_t^x := \{x_{t-1}, x_{t-2}, \dots\}$ and $\mathcal{I}_t^z := \{z_{t-1}, z_{t-2}, \dots\}$.
- Expanding the information set from $\mathcal{I}_t^x$ to $\mathcal{I}_t^x \cup \mathcal{I}_t^z$ is useful for forecasting $\{Y_t\}$.

Granger Causality: Testing for Granger Causality

- VAR(p) representation:

$$\begin{bmatrix} x_t \\ z_t \end{bmatrix} = \begin{bmatrix} \chi^x \\ \chi^z \end{bmatrix} + \begin{bmatrix} \phi_{11}^{(1)} & \phi_{12}^{(1)} \\ \phi_{21}^{(1)} & \phi_{22}^{(1)} \end{bmatrix} \begin{bmatrix} x_{t-1} \\ z_{t-1} \end{bmatrix} + \ldots + \begin{bmatrix} \phi_{11}^{(p)} & \phi_{12}^{(p)} \\ \phi_{21}^{(p)} & \phi_{22}^{(p)} \end{bmatrix} \begin{bmatrix} x_{t-p} \\ z_{t-p} \end{bmatrix} + \begin{bmatrix} \eta_t^x \\ \eta_t^z \end{bmatrix}$$

- The definition of Granger causality can be recast into $\{\phi_{12}^{(j)}\}_{j=1}^p$:
  - $\{z_t\}$ Granger-causes $\{x_t\}$ if $\phi_{12}^{(j)} \neq 0$ for some j;
  - $\{z_t\}$ does not Granger-cause $\{x_t\}$ if $\phi_{12}^{(j)} = 0$ for all j.
- The joint hypothesis to be tested:

$$\begin{cases} H_0 : \phi_{12}^{(1)} = \ldots = \phi_{12}^{(p)} = 0 \\ H_1 : \text{ not } H_0 \end{cases}$$

- This hypothesis can be tested by an F-test.

Granger Causality: Caveats

- The notion of “causality” or “causal effect” is not unique.
- Granger causality:
  - It is silent about the contemporaneous relationship between processes.
- *Ceteris paribus* causal effects (Marshall, 1890; Haavelmo, 1943, 1944):
  - The difference in the outcome across different policy regimes, with all else being equal.
  - e.g., elasticity, IRF, etc.
- Neyman-Rubin approach (Neyman, 1935; Rubin, 1972):
  - The treatment-control paradigm based on randomized controlled trials (RCTs).
  - e.g., average treatment effects (ATE), etc.
- Directed acyclic graphs (DAG; Pearl, 2009):
  - The causal relationship is represented by a network.

- Make sure that you are using an appropriate notion of “causality” or “causal effect”.

Summary and Preview

Summary

- We studied a multivariate time-series model for forecasting: VAR models!
  - VAR models account for intratemporal and intertemporal interactions.
- Two types of VAR(p) models:
  - Structural VAR
  - Reduced-form VAR
- Identification of the structural parameters requires additional assumptions.
- Granger causality

Preview

- In a finite sample, $\widehat{IRF}_t(h)$ is biased for $IRF_t(h)$.
- The bias tends to be large as h becomes large.
- Next week, we will study an alternative method to estimate $IRF_t(h)$ with a smaller bias.
  $\hookrightarrow$ Local projections!

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Stationarity Condition for VAR(p) Process

Proposition: Stability Condition for VAR(p) Process

The time-series process $\{Y_t\}$ with the reduce-form n-variate VAR(p) model

$$Y_t = B_0 + B_1 Y_{t-1} + B_2 Y_{t-2} + \ldots + B_{p-1} Y_{t-(p-1)} + B_p Y_{t-p} + \boldsymbol{\eta}_t,$$

is weakly stationary if and only if all the roots s^* of the corresponding characteristic equation

$$\det(\mathbf{I}_n - B_1 s - B_2 s^2 - \ldots - B_{p-1} s^{p-1} - B_p s^p) = 0$$

satisfy $|s^*| > 1$.

back

Stationarity Condition for VAR(p) Process

Sketch of Proof.

(Step 1) Companion form for the VAR(p) model: The reduced-form VAR(p) model can be transformed to

$$\tilde{Y}_t = C_0 + C_1 \tilde{Y}_{t-1} + \boldsymbol{\eta}_t,$$

where

$$\underset{np\times 1}{\tilde{Y}_t} := \begin{bmatrix} Y_t \\ Y_{t-1} \\ \vdots \\ Y_{t-(p-1)} \end{bmatrix}, \underset{np\times 1}{C_0} := \begin{bmatrix} B_0 \\ \mathbf{0} \\ \vdots \\ \mathbf{0} \end{bmatrix}, \underset{np\times np}{C_1} := \begin{bmatrix} B_1 & B_2 & \dots & B_{p-1} & B_p \\ \mathbf{I}_n & \mathbf{0} & \dots & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{I}_n & \ddots & \vdots & \mathbf{0} \\ \vdots & \ddots & \ddots & \mathbf{0} & \vdots \\ \mathbf{0} & \dots & \mathbf{0} & \mathbf{I}_n & \mathbf{0} \end{bmatrix}, \text{ and } \underset{np\times 1}{\tilde{\boldsymbol{\eta}}_t} := \begin{bmatrix} \boldsymbol{\eta}_t \\ \mathbf{0} \\ \vdots \\ \mathbf{0} \end{bmatrix}.$$

Stationarity Condition for VAR(p) Process

(Step 2) MA representation of the VAR(p) model: By repeated substitution,

$$\tilde{Y}_t = C_0 + C_1^t \tilde{Y}_0 + \sum_{\ell=1}^{t} C_1^{t-\ell} \tilde{\boldsymbol{\eta}}_{t-\ell}.$$

Stationarity Condition for VAR(p) Process

(Step 3) Eigenvalue decomposition: The reduced-form parameter matrix C_1 can be decomposed as

$$C_1 = \Gamma \Lambda \Gamma^{-1},$$

where

- $\underset{np \times np}{\Gamma}$: the matrix of eigenvectors;
- $\underset{np \times np}{\Lambda}$: the diaognal matrix of eigenvalues.

Stationarity Condition for VAR(p) Process

(Step 4) Stability:
Multiplying C_1 t times leads to

$$\begin{aligned} C_1^t &= C_1 C_1 C_1 \dots C_1 \\ &= \Gamma\Lambda\underbrace{\Gamma^{-1}\Gamma}_{\mathbf{I}_n}\Lambda\underbrace{\Gamma^{-1}\Gamma}_{\mathbf{I}_n}\Lambda\Gamma^{-1}\dots\Gamma\Lambda\Gamma^{-1} \\ &= \Gamma\Lambda^t\Gamma^{-1}. \end{aligned}$$

Let λ_{jj} be the jth diagonal element of Λ.
Observe that if $|\lambda_{jj}| \geq 1$ for some j, then every element of $\tilde{Y}_t$ goes to infinitly as $t \to \infty$, i.e., $E[Y_t]$ depends on time t.
Thus, for the process $\{Y_t\}$ to be independent of time t, it has to be that $|\lambda_{jj}| < 1$ for all j.

Stationarity Condition for VAR(p) Process

(Step 5) Characteristic equation:
Look at the characteristic equations:

$$
\begin{aligned}
&\det(\lambda \mathbf{I}_{np} - C_1) = 0 \\
&\iff s^{np}\det(\lambda \mathbf{I}_{np} - C_1) = 0 \\
&\iff \det(\mathbf{I}_{np} - sC_1) = 0 \\
&\iff \det(\mathbf{I}_{n} - B_1 s - B_2 s^2 - \ldots - B_{p-1}s^{p-1} - B_p s^p) = 0,
\end{aligned}
$$

where $s := \frac{1}{\lambda}$.
By definition, the eigenvalues $\{\lambda_{jj}\}$ are nothing but the solutions for this characteristic equation.

Stationarity Condition for VAR(p) Process

Hence, the stability condition reads: The VAR(p) process is weakly stationary if and only if all the roots λ^* for

$$\det(\lambda \mathbf{I}_{np} - C_1) = 0$$

satisfy $|\lambda| < 1$.
Equivalently, the VAR(p) process is weakly stationary if and only if all the roots s^* for

$$\det(\mathbf{I}_n - B_1 s - B_2 s^2 - \ldots - B_{p-1} s^{p-1} - B_p s^p) = 0$$

satisfy $|s^*| > 1$. $\square$

back